Personalized Learning Resource Recommendation Framework Based on Knowledge Graph and Large Language Model

Jianguo Chen[1,2] and Tongyu Zhu[1,2](✉)

[1] Hangzhou International Innovation Institute of Beihang University, Hangzhou, China
[2] School of Computer Science and Engineering, Beihang University, Beijing, China
{chenjianguo,zhutongyu}@buaa.edu.cn

Abstract. In the realm of digital education, Educational Recommender Systems (ERSs) play a pivotal role in enhancing learning outcomes. Nevertheless, existing systems encounter significant challenges, including data sparsity, cold-start problems, limited dynamic adaptability, insufficient personalization, and poor interpretability. To tackle these issues, we introduce an innovative personalized recommendation framework that integrates knowledge graphs (KGs) with Large Language Models (LLMs). By utilizing LLMs to enrich knowledge graphs—such as through the automatic generation of resource tags, difficulty levels, and overviews—our framework effectively addresses data sparsity and cold-start problems. It dynamically updates the knowledge graph and recommendation strategies based on real-time changes in students' knowledge states, while also enabling learners to adjust their knowledge profiles via Open Learner Models (OLMs). Our recommendation module synergizes knowledge graphs with advanced algorithms to compute the similarity between students and educational resources, facilitating personalized ranking. To enhance the interpretability of recommendations, our system extracts relevant subgraphs and articulates the recommendation rationale through natural language generation techniques. Empirical evidence demonstrates that this framework not only significantly boosts the quality and efficacy of recommendations but also fosters personalized education and elevates the intelligence of educational systems.

Keywords: Education Recommendation System · Knowledge Graph · Large Language Model · Open Learner Models · Bayesian Knowledge Tracing

1 Introduction

With the rapid development of online education platforms, Educational Recommender Systems (ERSs) have become essential for enhancing learning efficiency

© The Author(s), under exclusive license to Springer Nature Singapore Pte Ltd. 2026
T. Zhu et al. (Eds.): KSEM 2025, LNAI 15919, pp. 228–245, 2026.
https://doi.org/10.1007/978-981-95-3001-4_17

and personalization. However, existing systems face challenges in dynamic adaptability, interpretability, and data sparsity. Traditional recommendation methods, such as collaborative filtering [20] and content-based filtering [33], struggle with cold-start issues and fail to capture the evolving knowledge states of learners. Although Knowledge Graph (KG)-based systems improve interpretability [3,27], their static nature limits real-time adaptability. Meanwhile, Large Language Models (LLMs) offer new possibilities for semantic analysis and feature generation [8], yet their integration with dynamic knowledge modeling remains an open question. The primary challenges in current ERS research are as follows:

- Data sparsity and cold-start issues: Limited resource metadata hampers accurate user profiling, reducing recommendation quality.
- Static knowledge representations: Existing models fail to dynamically track and adapt to learners' evolving knowledge states.
- Lack of interpretability: The "black-box" nature of many recommendation processes undermines user trust and usability [16].
- Imbalance between personalization and learning goals: Current approaches struggle to align recommendations with both knowledge gaps and diverse learning interests.

To address these challenges, this paper proposes a personalized learning resource recommendation framework integrating knowledge graphs and LLMs. The key innovations include:

- Dynamic Knowledge Enhancement: By leveraging LLMs to generate multimodal resource features, addressing cold-start and data sparsity issues.
- Transparent Learning Modeling: The integration of Bayesian Knowledge Tracing (BKT) [35] and Open Learner Models (OLMs) [11] allows users to participate in knowledge state adjustments, enhancing recommendation interpretability and personalization.
- Efficient Hybrid Recommendation Algorithm: A scoring function based on semantic alignment, knowledge gaps, and difficulty stratification is proposed, combined with reinforcement learning and indexing optimization techniques to achieve low-latency, high-precision recommendations.

Experiments demonstrate that the proposed framework achieves significant improvements in recommendation accuracy and recall across multiple datasets, including ASSISTments2012 [25], POJ [25], and BUAA-OODC, outperforming mainstream baseline models such as KG-RS and CF+KG-RS. The framework's effectiveness is further corroborated through semantic relevance evaluation using ROUGE-L [22] and diversity assessment via Intra-List Diversity (ILD) [18]. Ablation studies confirm the critical contributions of both the LLM-driven data augmentation strategy and the OLM module to the overall performance enhancement. Notably, empirical teaching data reveal that the framework's deployment correlates with a marked improvement in students' homework completion outcomes compared to historical baselines, demonstrating its practical efficacy in educational environments.

The remainder of this paper is organized as follows: Sect. 2 reviews related work; Sect. 3 details the framework design and algorithm implementation; Sect. 4 validates the framework's effectiveness through experiments; and Sect. 5 concludes the paper and outlines future directions.

2 Related Work

2.1 Learner Modeling in Educational Recommender Systems

Learner models are essential in Educational Recommender Systems (ERSs), providing an abstract representation of learners' abilities and behaviors by analyzing their interactions with the system [12]. A key component of the learner model is the learner's knowledge state [10]. Current approaches to learner modeling, particularly in ERSs, typically use "overlaying" models that estimate a student's knowledge based on system interactions [10]. Methods such as the Elo rating system [1], Performance Factor Analysis (PFA) [32], Bayesian Knowledge Tracing (BKT) [30], and Item Response Theory (IRT) [21] are commonly employed to generate these models. However, these methods often remain opaque, which may raise concerns regarding fairness and trust, as learners cannot easily understand how their knowledge state is inferred [2].

2.2 Content-Based Educational Recommendation System

In general, content-based ERSs recommend items similar to those the user has previously liked based on content features. These systems analyze the items rated by a user and construct a model to capture the user's interests, which is then used to recommend new items [26]. A recent survey on ERSs in MOOCs found that 26% of these systems are designed to recommend learning elements, including learning activities, video lectures, next pages, learning sources, and learning paths [19]. The authors noted that most research in this field focuses on content - based filtering or hybrid algorithms. Traditionally, content - based ERSs base their recommendations on content similarity [7], suggesting content similar to what learners have interacted with in the past, such as learning materials they have viewed or courses they have taken.

2.3 Education Recommendation System Based on Knowledge Graph

In recent years, researchers have explored the integration of knowledge graphs (KGs) into recommender systems (RS) to enhance recommendation accuracy and interpretability [17]. KGs represent entities as nodes and relationships as edges, allowing for a more comprehensive understanding of item relationships and user preferences when user information is incorporated [37]. While KG-based RS have been widely applied in domains such as tourism, health, entertainment, and business, their application in education remains limited. Few studies have proposed KG-based educational recommender systems (ERS) for recommending

learning paths [38] or labs [13]. Notably, the work in [31] introduced a KG-based approach for explainable recommendations of Wikipedia articles based on learner needs. However, this method relied on full-text search algorithms (e.g., Lucene), which, despite their efficiency, struggle to capture semantic similarities between texts with differing vocabularies.

2.4 LLM in Recommender Systems

With the advancements of Large Language Models (LLMs) [9], researchers have explored their use in recommendation tasks through two main approaches: (1) as predictors, directly generating recommendations [5,23,36]; (2) as extractors, creating user/item profiles to support downstream tasks [14,24]. Predictor-based methods like TALLRec (Task-Aware Learning for Recommender Systems) [5] fine-tune LLMs with recommendation data, addressing task mismatches and data scarcity. Extractor-based methods, such as P5 (Personalized Prediction for Recommendation Systems) [14], unify diverse data into natural language for instruction-based recommendations, enhancing personalization. Additionally, LLMRec (Large Language Model-based Recommender Systems) [34] and RLMRec (Reinforcement Learning-based Model for Recommender Systems) [29] integrate LLMs with GNNs (Graph Neural Networks) to improve representation performance and mitigate data sparsity.

2.5 Summary of Contributions

In summary, while existing research has independently applied learner modeling, knowledge graphs, and LLMs to educational recommenders, these approaches often suffer from being static, opaque, or semantically limited. Our framework introduces a novel, synergistic integration of these components to overcome these specific limitations. The key breakthroughs of our work are: (1) Employing LLMs for the dynamic enrichment of the knowledge graph, directly addressing data sparsity and improving semantic understanding beyond traditional content-based methods. (2) Fusing Bayesian Knowledge Tracing with Open Learner Models to create a transparent and interactive learner profile, moving beyond the "black-box" nature of conventional learner models. (3) Establishing a dynamic feedback loop where the learner's evolving knowledge state continuously updates the graph, enabling adaptive and timely recommendations that static KG-based systems cannot provide. This integrated approach marks a significant step towards a more personalized, interpretable, and effective educational recommendation system.

3 Methodology

This study introduces a personalized learning resource recommendation framework that integrates knowledge graphs (KGs) with Large Language Models (LLMs), as illustrated in Fig. 1. Initially, the framework gathers online learning

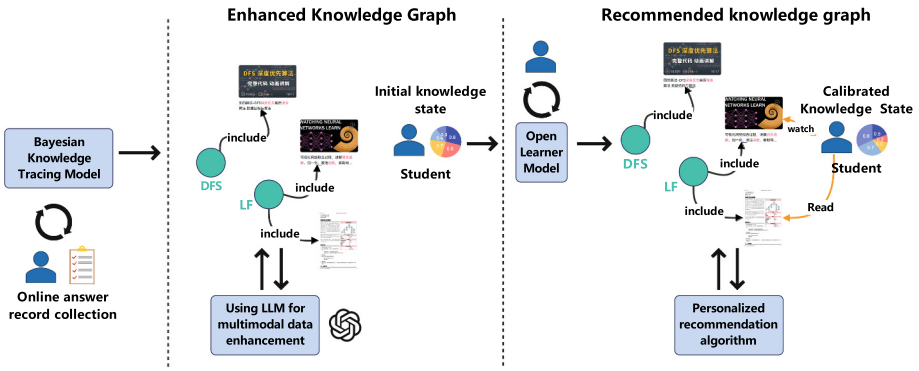


Fig. 1. This figure shows the workflow and architecture of the personalized learning resource recommendation framework based on knowledge graph and large language model.

data from students and employs Bayesian Knowledge Tracing (BKT) to assess students' mastery of various topics, thereby establishing their initial knowledge states. Based on this information, a knowledge graph is constructed, featuring nodes representing students, knowledge domains, and learning resources. To enrich this graph, LLMs are utilized to enhance resource nodes by generating tags, categorizing difficulty levels, and producing summaries, thus providing more comprehensive resource descriptions. Furthermore, the OLM allows students to manually adjust their initial knowledge states, enabling a higher degree of personalization. Finally, advanced recommendation algorithms analyze the relationships between student and resource nodes to suggest tailored learning materials. These recommendations are visually represented within the graph, enhancing both the interpretability and effectiveness of the recommendation outcomes.

3.1 Knowledge Graph Construction and Multimodal Data Augmentation

Construction of Knowledge Graph. The BKT model is employed to quantify the degree of mastery of knowledge points for each student. This model generates a personalized knowledge status table, which represents the student's knowledge proficiency on a scale from 0 (not mastered) to 1 (fully mastered). To support this framework, we constructed a basic knowledge graph using Neo4j, comprising student nodes, knowledge nodes, and resource nodes. In this graph, knowledge nodes are connected to resource nodes via an "include" relationship, while student nodes are associated with resource nodes through behaviors such as "watching" or "reading." Each student node includes attributes such as knowledge status, ID, gender, and instructor, whereas resource nodes record information about resource types and URLs. For example, the basic knowledge graph depicted in Fig. 1 encompasses two knowledge nodes ("depth-first search"

and "loss function"), three resource nodes (in both text and video formats), and one student node.

Multimodal Data Augmentation. To further enrich the content of the basic knowledge graph, we leverage LLM to conduct in-depth analysis of the resources, thereby enhancing the data. As depicted in Fig. 2, for video resources, in addition to evaluating basic information such as titles and introductions, we extract and convert video subtitles into text format. The LLM then analyzes this text information to generate three features for video resources: content labels, content overviews, and difficulty levels. For text resources, the LLM directly generates these three features—labels, content overviews, and difficulty levels—based on the analysis of the text content. Finally, these features are integrated into the knowledge graph to form an enhanced knowledge graph. In this enhanced graph, the attributes of resource nodes are expanded from the original two (resource type and resource URL) to five, with the addition of three new features: resource labels, resource difficulty, and resource overview. This approach effectively increases the information density and application value of the knowledge graph.

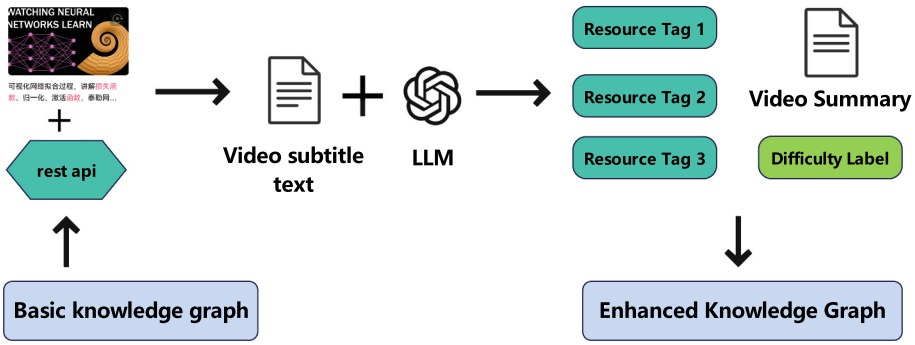


Fig. 2. The diagram illustrates the process of enhancing knowledge graphs using LLMs.

3.2 Knowledge State Modeling

In this framework, we integrate the strengths of closed learner models (e.g., BKT) and open learner models (OLMs) to model student knowledge states, aiming to combine the objectivity of the former with the interpretability of the latter. The OLM, as an externalized student model, is accessible to both students and teachers [11] and has been incorporated into ERS to enhance the transparency and interpretability of these systems [1,2,6]. As illustrated in Fig. 3, each student can visualize their initial knowledge mastery status calculated by the BKT model, and crucially, may manually adjust their self-perceived mastery levels

of specific knowledge components based on their actual learning progress. This explicit feedback mechanism aligns with studies emphasizing the importance of improving user model auditability in recommender systems [4,15]. Through the OLM interface, students' adjustments transform the initial knowledge states into revised ones, which are subsequently integrated into the enhanced knowledge graph. This dual-process approach preserves Bayesian objectivity while enabling personalized calibration, thereby enhancing both individual adaptability and system transparency throughout the learning process.

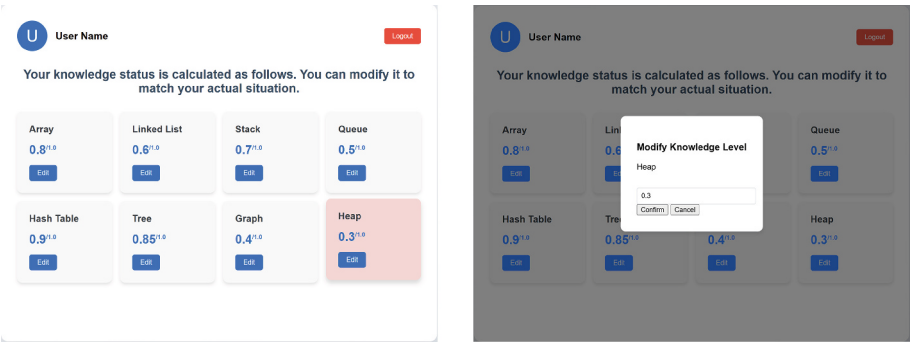


Fig. 3. This diagram presents the user interface of our OLM. The model empowers students to modify system representations to align with their actual situations, facilitating a more personalized learning experience.

3.3 Personalized Recommendation Algorithm

This section presents a dynamic recommendation algorithm combining three key innovations: 1) BKT-based knowledge tracing enhanced with student-adjusted OLM refinement, 2) Multi-level difficulty matching (beginner/intermediate/advanced) through ZPD-constrained scoring, and 3) BERT-Faiss semantic alignment accelerated via HSW indexing. The hybrid approach addresses educational recommendation challenges through temporal student modeling, reinforcement learning-optimized weights, and cold-start handling via knowledge gap propagation, overcoming static feature limitations in conventional methods.

Problem Formalization. The heterogeneous knowledge graph comprises three entity types:

– **Student nodes** s: Represent learners with dynamically updated knowledge states $K_s \in [0, 1]$ calculated via BKT:

$$K_s^{(t)} = P(L_t = 1|O_{1:t-1}) = \frac{P(O_{t-1}|L_{t-1} = 1)P(L_{t-1} = 1)}{\sum_l P(O_{t-1}|L_{t-1} = l)P(L_{t-1} = l)}, \quad (1)$$

where L_t denotes latent mastery states and O_t represents observed performance. The final knowledge mastery state $K_s^{(t)'}$ is obtained by applying the OLM model to $K_s^{(t)}$:

$$K_s^{(t)'} = \mathrm{OLM}(K_s^{(t)}, \Delta_s), \tag{2}$$

where Δ_s represents the manual adjustment made by the student. The OLM model allows students to fine-tune their knowledge mastery based on self-assessment, providing adjustments to the BKT-calculated mastery to better reflect their actual understanding.

- **Knowledge nodes** k: Construct hierarchical dependencies from curriculum standards.
- **Resource nodes** r_i: Characterized by tuple

$$\mathbf{R_i} = \big(\text{type}, \mathcal{T}_i, \text{url}, d_i, \text{desc}\big), \tag{3}$$

where $d_i \in \{\text{beginner}, \text{intermediate}, \text{advanced}\}$ and $\mathcal{T}_i$ denotes standardized semantic tags.

Scoring Function. The recommendation score synthesizes three pedagogically motivated components:

$$S_i = w_1(1 - \bar{K}'_s) + w_2 \mathrm{sim}_{\mathrm{BERT}}(\mathcal{T}_s, \mathcal{T}_i) + w_3 \Phi(K'_s, d_i), \tag{4}$$

The **knowledge gap term** $1 - \bar{K}'_s$ quantifies unmastered content using the average OLM-adjusted knowledge mastery over T interactions. The **semantic alignment term** employs pre-trained BERT embeddings to capture deep tag relationships:

$$\mathrm{sim}_{\mathrm{BERT}}(\mathcal{T}_s, \mathcal{T}_i) = \cos\left(\frac{1}{|\mathcal{T}_s|}\sum_{t\in\mathcal{T}_s}\mathbf{v}_t, \frac{1}{|\mathcal{T}_i|}\sum_{t\in\mathcal{T}_i}\mathbf{v}_t\right), \tag{5}$$

where $\mathbf{v}_t$ denotes the BERT [CLS] embedding of tag t. The **difficulty matching term** $\Phi(K'_s, d_i)$ enforces zone of proximal development principles:

$$\Phi(K'_s, d_i) = \begin{cases} 1, & K'_s < 0.4 \wedge d_i = \text{beginner} \\ 0.5, & 0.4 \le K'_s < 0.7 \wedge d_i = \text{intermediate} \\ 0.2, & K'_s \ge 0.7 \wedge d_i = \text{advanced}. \end{cases} \tag{6}$$

The algorithm implements three key optimizations: (1) **Reinforcement learning-based dynamic weight adjustment** optimizes recommendation weights adaptively using NDCG feedback; (2) **Faiss-based hierarchical small-world (HSW) indexing** accelerates semantic retrieval, achieving a response time constraint of 300ms for 10K resources; (3) **Cold-start handling via knowledge gap-based collaborative filtering**, ensuring recommendations even when student history is unavailable.

Dynamic Weight Adjustment via Reinforcement Learning. To optimize the recommendation score function adaptively, we update the weight parameters w_j based on NDCG feedback:

$$w_j^{(t+1)} = w_j^{(t)} + \eta \cdot \text{NDCG@}k \cdot \frac{\partial S}{\partial w_j}, \tag{7}$$

where η is the learning rate, and NDCG@k reflects ranking quality. This enables the system to self-optimize without manual tuning.

Semantic Computation Acceleration. We accelerate semantic similarity search using **Faiss-based hierarchical small-world (HSW) indexing**, achieving real-time retrieval with:

$$T_{\text{response}} < 300\text{ms}, \quad \mathcal{R}_{\text{ranked}} = \underset{r_j \in \mathcal{R}}{\text{argtopk}} \left(S_j - \max_{r_i \in \mathcal{R}_{\text{selected}}} \text{sim}(r_i, r_j) \right), \tag{8}$$

This balances relevance and diversity while ensuring low latency.

Cold-Start Handling via Knowledge Gap-Based Filtering. When no prior student data is available ($T_s = \emptyset$), recommendations are based on knowledge gaps:

$$S_i^{\text{cold}} = w_1(1 - K_s) + w_3 \text{diff}(K_s, d_i), \tag{9}$$

where K_s is the student's knowledge proficiency, and d_i is resource difficulty. This ensures meaningful recommendations even for new users.

Algorithm Implementation. The recommendation process formalizes as follows:

Parameter Selection and Tuning

- **BKT**: The parameters of the BKT model are not pre-set, but estimated from the data through the expectation maximization (EM) [28] algorithm. In order to overcome the local optimal problem that may be caused by the sensitivity of the EM algorithm to the initial value, a multi-starting point initialization strategy is adopted: the model is randomly initialized multiple times (5 times by default) and the EM algorithm is run until convergence, and finally a set of parameters with the largest log-likelihood value is selected as the estimation result.
- **RL Weights**: To determine the optimal hyperparameter combination, including the learning rate η and k in NDCG@k, we employed a grid search approach on the validation set for tuning. For the learning rate η, we tested candidate values from $\{0.001, 0.01, 0.05, 0.1\}$, where a too-small η could lead to slow convergence, while an excessively large η might cause weights to oscillate or even diverge near the optimum. For k in NDCG@k, we evaluated

Algorithm 1. Personalized Educational Resource Recommendation

Require: Student s, knowledge graph $\mathcal{G}$, top-k resources
Ensure: Recommended resources $\mathcal{R}_{\text{rec}}$

1: Identify weak knowledge points: $\mathcal{K} \leftarrow \{k \mid \bar{K}_s'^k < 0.4, \forall k \in \mathcal{G}\}$
2: Retrieve candidate resources: $\mathcal{R} \leftarrow \bigcup_{k \in \mathcal{K}} \mathcal{G}.\text{getResources}(k)$
3: Filter by difficulty: $\mathcal{R}' \leftarrow \{r \in \mathcal{R} \mid \Phi(K_s'^{k_r}, d_r) > 0\}$
4: Compute BERT embeddings: $\mathbf{V} \leftarrow \text{Faiss-HSW}(\{\mathcal{T}_i \mid r_i \in \mathcal{R}'\})$ ▷ Faiss-based accelerated computation
5: Calculate similarity scores: $\text{sim}_i \leftarrow \cos(\mathbf{v}_s, \mathbf{v}_i), \forall \mathbf{v}_i \in \mathbf{V}$
6: Generate final scores:

$$\begin{cases} S_i \leftarrow w_1(1 - \bar{K}_s') + w_2 \text{sim}_i + w_3 \Phi(K_s', d_i), & \text{if } T_s \neq \emptyset \\ S_i^{\text{cold}} \leftarrow w_1(1 - K_s) + w_3 \text{diff}(K_s, d_i), & \text{if } T_s = \emptyset \end{cases}$$

7: Diversify results: $\mathcal{R}_{\text{ranked}} \leftarrow \text{NMS}(\mathcal{R}', S, \lambda = 0.6)$
8: Select top-k: $\mathcal{R}_{\text{rec}} \leftarrow \mathcal{R}_{\text{ranked}}[1:k]$

$k \in \{5, 10, 20\}$ based on typical user interface display quantities, as the choice of k influences the focus range of reward signals. Our tuning objective was to identify a pair (η, k) that maximizes the average NDCG@k performance on the validation set. The experimental procedure involved iteratively training the model on the training set for each hyperparameter combination and evaluating its performance on the validation set after each epoch. Ultimately, we selected $\eta = 0.01$ and $k = 5$ as the optimal hyperparameters, as this combination achieved the highest and most stable average NDCG on the validation set.

4 Experiments

4.1 Datasets and Baselines

Datasets. We used the following datasets to evaluate the proposed framework:

- **ASSISTments2012:** This is the ASSISTments data for the school year 2012 2013 with affect predictions. The dataset consists of 2,541,201 interactions, 27,066 students, and 45,716 questions.
- **POJ:** This dataset consists of programming exercises and is collected from Peking coding practice online platform. The dataset is originally scraped by Pandey and Srivastava. In total, it has 996,240 interactions, 22,916 students, and 2,750 questions.
- **BUAA-OODC:** Collected from the Object-oriented design and construction course (OODC) at Beihang University's School of Computer Science and Engineering, this dataset spans five academic years (2019–2024) and includes students' online quiz records.

Baselines. To evaluate the effectiveness of our proposed framework, we compared it with the following baseline models:

- **Knowledge Graph-Based Recommendation (KG-RS):** This model incorporates a static knowledge graph (KG) to recommend resources based on the relationships and attributes of entities within the graph. By mapping users, resources, and knowledge points into the graph, it leverages the connections between these entities to enhance recommendation accuracy.
- **Collaborative Filtering + Knowledge Graph-Based Recommendation (CF+KG-RS):** This model combines collaborative filtering (CF) with the knowledge graph (KG) to recommend resources. It uses the interaction patterns of similar users and the structured information from the KG to make recommendations for a target user.
- **Content-Based Filtering + Knowledge Graph-Based Recommendation (CBF+KG-RS):** This model enhances content-based filtering (CBF) with a knowledge graph (KG). It uses the content features of resources and the KG to recommend resources that are similar to what the user has interacted with before, based on both content similarity and knowledge relationships.

4.2 Experimental Setup and Evaluation Metrics

Experimental Setup. In our experimental setup, we first conduct data preprocessing, which includes cleaning and normalizing the data to handle missing values and eliminate duplicates. The dataset is then partitioned into three subsets: 70% for training, 15% for validation, and 15% for testing. For knowledge graph construction, we use the Bayesian Knowledge Tracing (BKT) model to initialize student knowledge states and employ an Large Language Model (LLM) to generate labels, summaries, and difficulty levels to enrich crawled resource nodes. In the parameter tuning phase, we apply an adaptive weight optimization method based on reinforcement learning to optimize the scoring function's weight parameters. For implementation, we build the recommendation algorithm using PyTorch and utilize the GPT-4o LLM for data augmentation and explanation generation. Note that the Open Learner Model (OLM) module, requiring verification in a real educational scenario, is applied only to our established BUAA-OODC dataset.

Evaluation Metrics. We evaluate our recommendation system using five metrics: Precision@k, Recall@k, HR@k, ILD (Intra-List Diversity), and ROUGE-L. These metrics assess recommendation accuracy, coverage, personalization, and interpretability.

4.3 Experimental Results

The experimental results demonstrate that our proposed framework significantly outperforms baseline models across three datasets: ASSISTments2012, POJ, and

BUAA-OODC. As shown in Tables 1, when compared to knowledge graph (KG)-based recommendation systems, collaborative filtering plus KG-based systems, and content-based filtering plus KG-based systems, our framework consistently achieves higher scores in Precision@k, Recall@k, and Hit Rate@k (HR@k). Additionally, it delivers superior diversity (measured by Intra-List Diversity, ILD) and content relevance (assessed via ROUGE-L). On the ASSISTments2012 dataset, our framework achieved precision and recall improvements of up to 21.2% and 20.0%, respectively, compared to the KG-based recommendation system. This highlights its efficacy in providing more accurate and diverse recommendations. On the POJ dataset, similar gains were observed, with recall increasing by 19.4% over the KG-based model. Finally, on the BUAA-OODC dataset, our framework attained a precision of 0.83 and an HR@k of 0.84, signifying substantial enhancements across all evaluation metrics. These results underscore the robustness and adaptability of our proposed recommendation framework. It effectively captures user preferences and provides more personalized and diverse learning resource recommendations compared to traditional methods.

Table 1. Performance Comparison of Different Models on Various Datasets

Dataset	Model	Precision@k	Recall@k	HR@k	ILD	ROUGE-L
ASSISTments2012	KG	0.67	0.60	0.68	0.57	0.61
	CF+KG	0.76	0.64	0.72	0.76	0.74
	CBF+KG	0.79	0.68	0.75	0.72	0.73
	Ours	0.85	0.75	0.79	0.81	0.80
POJ	KG	0.65	0.58	0.63	0.61	0.66
	CF+KG	0.77	0.65	0.80	0.77	0.73
	CBF+KG	0.74	0.69	0.74	0.68	0.76
	Ours	0.83	0.72	0.81	0.78	0.85
BUAA-OODC	KG	0.61	0.62	0.59	0.65	0.68
	CF+KG	0.73	0.66	0.74	0.64	0.75
	CBF+KG	0.78	0.70	0.77	0.81	0.74
	Ours	0.83	0.78	0.84	0.82	0.83

4.4 Ablation Study

To validate the contributions of LLM-based data augmentation and the OLM, we conducted ablation studies on three datasets: ASSISTments2012, POJ, and BUAA-OODC2019-2024. Specifically, we tested two variants: (1) Ours w/o LLM, where resource tags, summaries, and difficulty levels rely solely on raw metadata, and (2) Ours w/o OLM, where knowledge states are updated only via BKT without manual adjustments. Performance was evaluated using metrics such as

Precision@10, Coverage@20, Knowledge Gain Rate (KGR), Dynamic Adaptation Measure (DAM), and Cold-Start Rate (CSR). The results are summarized in Tables 2, 3, and 4.

Table 2. Ablation Study Results on ASSISTments2012 Dataset

Model Variant	Precision@10	Recall@10	HR@10	ILD	ROUGE-L
Full Model (Ours)	0.85	0.75	0.79	0.81	0.80
Ours w/o LLM	0.67 (−21.2%)	0.60 (−20.0%)	0.68 (−13.9%)	0.57 (−29.6%)	0.61 (−23.8%)
Ours w/o OLM	0.85 (0.0%)	0.75 (0.0%)	0.79 (0.0%)	0.81 (0.0%)	0.80 (0.0%)

Table 3. Ablation Study Results on POJ Dataset

Model Variant	Precision@10	Recall@10	HR@10	ILD	ROUGE-L
Full Model (Ours)	0.83	0.72	0.81	0.78	0.85
Ours w/o LLM	0.65 (−21.7%)	0.58 (−19.4%)	0.63 (−22.2%)	0.61 (−21.8%)	0.66 (−22.4%)
Ours w/o OLM	0.83 (0.0%)	0.72 (0.0%)	0.81 (0.0%)	0.78 (0.0%)	0.85 (0.0%)

Table 4. Ablation Study Results on BUAA-OODC Dataset

Model Variant	Precision@10	Recall@10	HR@10	ILD	ROUGE-L
Full Model (Ours)	0.83	0.78	0.84	0.82	0.83
Ours w/o LLM	0.61 (−26.5%)	0.62 (−20.5%)	0.59 (−29.8%)	0.65 (−20.7%)	0.68 (−18.1%)
Ours w/o OLM	0.82 (−1.2%)	0.79 (+1.3%)	0.82 (−2.4%)	0.70 (−14.6%)	0.73 (−12.0%)

Key Findings and Statistical Significance. Our ablation studies highlight the critical role of LLM-based data augmentation and the necessity of the OLM:

- **LLM's Impact**: Across all datasets, disabling the LLM-based augmentation leads to a significant performance drop. For instance, on the ASSISTments2012 dataset, Precision@10 decreased from 0.85 to 0.67 (a reduction of 21.2%), Recall@10 dropped by 20.0%, and HR@10 fell by 13.9%, while ILD and ROUGE-L declined by 29.6% and 23.8% respectively. Similar trends were observed on the POJ and BUAA-OODC datasets, with Precision@10 on BUAA-OODC decreasing by up to 26.5%. These results indicate that the LLM module effectively mitigates the sparsity of raw metadata by generating enriched resource tags, summaries, and difficulty levels, thereby substantially enhancing overall system performance.

– **OLM's Impact**: The necessity of the OLM module hinges on real-time user-system interactions. Experiments reveal distinct outcomes: On static datasets (ASSISTments2012 and POJ, lacking iterative interactions), removing OLM caused marginal performance differences, indicating that baseline knowledge tracing (e.g., BKT) suffices for modeling user states. In contrast, on BUAA-OODC—a dynamic interaction dataset with authentic user feedback—disabling OLM led to significant declines: Precision@10 (−1.2%), HR@10 (−2.4%), ILD (−14.6%), and ROUGE-L (−12.0%). This stark contrast underscores OLM's critical role in dynamically adapting to user behavior in interactive recommendation scenarios.

Discussion. Our ablation studies reinforce the value of LLM-based data augmentation and the OLM. The LLM effectively addresses resource metadata sparsity by generating semantic tags and difficulty levels, leading to better personalization and diversity. Meanwhile, the OLM improves adaptability through user-driven knowledge state updates. However, LLM inference introduces an approximate 15% latency increase, necessitating future research on optimizing lightweight deployment strategies.

4.5 User Study and Visualization

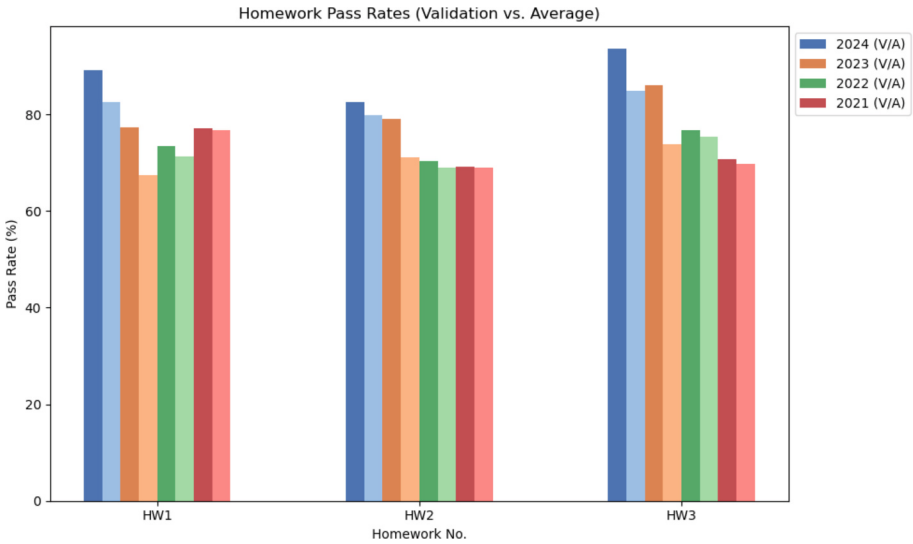
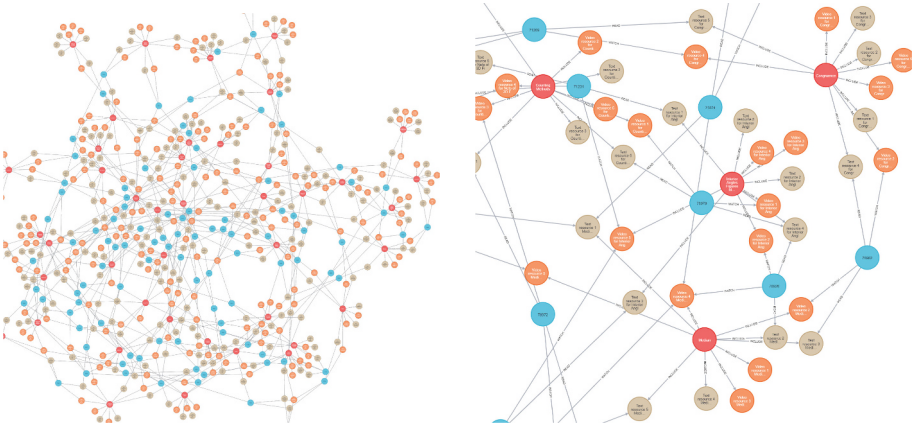


Fig. 4. The effect diagram of teaching measures integrating the personalized learning resource recommendation framework of this paper.



(a) (b)

Fig. 5. Partial display of the knowledge graph based on the ASSISTments2012 dataset: (a) shows a thumbnail of a local area, and for clearer observation, a part of (a) is enlarged in (b).

User Study. In alignment with the teaching reform initiative at Beihang University, specifically the project titled "Construction and Analysis of Student Learning Portraits for Software Competence Teaching Process," the project leader implemented a learning resource recommendation framework in the 2023 and 2024 teaching cohorts for the courses under their purview. As depicted in Fig. 4, prior to the introduction of this framework in 2021 and 2022, the homework pass rates of these cohorts were comparable to the average levels. However, following the adoption of the framework in 2023 and 2024, the homework pass rates significantly surpassed the average levels. This evidence underscores the efficacy of personalized learning resource recommendations in enhancing students' knowledge acquisition and comprehension. Moreover, it reaffirms the critical role of targeted learning support within the teaching process.

Visualization. Taking the ASSISTments2012 dataset as an example, a subset comprising 5,000 student nodes and 100 skill nodes was extracted. To further construct an educational resource map, relevant educational resources corresponding to the aforementioned skill names were collected from the Wikipedia and Bilibili platforms. Following identification and screening, a total of 1,251 pertinent resources were obtained. These resources consist of 475 textual resources (Wikipedia content) and 776 video resources (Bilibili content).

Based on the three entity types—students, skills, and resources—a knowledge graph structure was constructed, resulting in tens of thousands of data triples. Specifically, the graph includes: Skill-resource relationships: 1,251 association relationships (indicating that a skill possesses corresponding teaching resources). Student-resource interaction relationships: 21,744 learning interaction records,

comprising 7,980 reading interactions (corresponding to text resources) and 13,764 viewing interactions (corresponding to video resources). Finally, the knowledge graph data was stored within the Neo4j graph database. The visualization of the resultant knowledge graph is presented in Fig. 5.

5 Conclusion

Our paper proposed a new personalized learning resource recommendation framework combining knowledge graphs (KGs) and Large Language Models (LLMs) to tackle key challenges in Educational Recommender System (ERS). Experiments showed its superiority over other methods. It greatly improved Precision@10, Recall@10, diversity and semantic relevance. Ablation studies highlighted the importance of LLM - driven data augmentation and personalization enabled by Open Learner Model (OLM). Practical use in classrooms also proved its effectiveness, with increased homework pass rates at Beihang University.

However, there are still limitations. LLM inference causes extra computational latency, and the current framework mainly works for structured knowledge areas. Future work will look into federated learning for data privacy, multimodal learner behavior analysis, cross - domain knowledge transfer, as well as supporting collaborative and lifelong learning. This study offers a scalable and interpretable solution for intelligent education systems and provides insights and practical guidance for personalized education in the age of generative artificial intelligence.

Acknowledgements. This work was supported by the National Natural Science Foundation of China (U24B20171,62394332) and Research Start-up Funds of Hangzhou International Innovation Institute of Beihang University (2024KQ068, 2024KQ053).

References

1. Abdi, S., Khosravi, H., Sadiq, S., Gasevic, D.: A multivariate ELO-based learner model for adaptive educational systems. arXiv preprint arXiv:1910.12581 (2019)
2. Abdi, S., Khosravi, H., Sadiq, S., Gasevic, D.: Complementing educational recommender systems with open learner models. In: Proceedings of the Tenth International Conference on Learning Analytics & Knowledge, pp. 360–365 (2020)
3. Ain, Q.U., Chatti, M.A., Meteng Kamdem, P.A., Alatrash, R., Joarder, S., Siepmann, C.: Learner modeling and recommendation of learning resources using personal knowledge graphs. In: Proceedings of the 14th Learning Analytics and Knowledge Conference, pp. 273–283 (2024)
4. Balog, K., Radlinski, F., Arakelyan, S.: Transparent, scrutable and explainable user models for personalized recommendation. In: Proceedings of the 42nd International ACM SIGIR Conference on Research and Development in Information Retrieval, pp. 265–274 (2019)
5. Bao, K., Zhang, J., Zhang, Y., Wang, W., Feng, F., He, X.: Tallrec: an effective and efficient tuning framework to align large language model with recommendation. In: Proceedings of the 17th ACM Conference on Recommender Systems, pp. 1007–1014 (2023)

6. Barria-Pineda, J., Brusilovsky, P.: Explaining educational recommendations through a concept-level knowledge visualization. In: Companion Proceedings of the 24th International Conference on Intelligent User Interfaces, pp. 103–104 (2019)
7. Benedict, A., Al-Hossami, E., Dorodchi, M., Benedict, A., Wiktor, S.: Pilot recommender system enabling students to indirectly help each other and foster belonging through reflections. In: LAK22: 12th International Learning Analytics and Knowledge Conference, pp. 521–527 (2022)
8. Bonifacio, L., Abonizio, H., Fadaee, M., Nogueira, R.: InPars: data augmentation for information retrieval using large language models. arXiv preprint arXiv:2202.05144 (2022)
9. Brown, T., et al.: Language models are few-shot learners. Adv. Neural. Inf. Process. Syst. **33**, 1877–1901 (2020)
10. Brusilovsky, P., Millán, E.: User models for adaptive hypermedia and adaptive educational systems. In: Brusilovsky, P., Kobsa, A., Nejdl, W. (eds.) The Adaptive Web. LNCS, vol. 4321, pp. 3–53. Springer, Heidelberg (2007). https://doi.org/10.1007/978-3-540-72079-9_1
11. Bull, S., Kay, J.: SMILI: a framework for interfaces to learning data in open learner models, learning analytics and related fields. Int. J. Artif. Intell. Educ. **26**, 293–331 (2016)
12. Conati, C., Porayska-Pomsta, K., Mavrikis, M.: AI in education needs interpretable machine learning: Lessons from open learner modelling. arXiv preprint arXiv:1807.00154 (2018)
13. Deng, Y., Lu, D., Huang, D., Chung, C.J., Lin, F.: Knowledge graph based learning guidance for cybersecurity hands-on labs. In: Proceedings of the ACM Conference on Global Computing Education, pp. 194–200 (2019)
14. Geng, S., Liu, S., Fu, Z., Ge, Y., Zhang, Y.: Recommendation as language processing (RLP): a unified pretrain, personalized prompt & predict paradigm (p5). In: Proceedings of the 16th ACM Conference on Recommender Systems, pp. 299–315 (2022)
15. Guesmi, M., et al.: Open, scrutable and explainable interest models for transparent recommendation. In: IUI Workshops (2021)
16. Guidotti, R., Monreale, A., Ruggieri, S., Turini, F., Giannotti, F., Pedreschi, D.: A survey of methods for explaining black box models. ACM Comput. Surv. (CSUR) **51**(5), 1–42 (2018)
17. Guo, Q., et al.: A survey on knowledge graph-based recommender systems. IEEE Trans. Knowl. Data Eng. **34**(8), 3549–3568 (2020)
18. Jesse, M., Bauer, C., Jannach, D.: Intra-list similarity and human diversity perceptions of recommendations: the details matter. User Model. User-Adap. Inter. **33**(4), 769–802 (2023)
19. Khalid, A., Lundqvist, K., Yates, A.: Recommender systems for MOOCS: a systematic literature survey (January 1, 2012-July 12, 2019). Int. Rev. Res. Open Distrib. Learn. **21**(4), 255–291 (2020)
20. Koren, Y., Rendle, S., Bell, R.: Advances in collaborative filtering. Recommender Systems Handbook, pp. 91–142 (2021)
21. Leite, L., et al.: A novel video recommendation system for algebra: an effectiveness evaluation study. In: LAK22: 12th International Learning Analytics and Knowledge Conference, pp. 294–303 (2022)
22. Lin, C.Y.: Rouge: a package for automatic evaluation of summaries. In: Text Summarization Branches Out, pp. 74–81 (2004)

23. Liu, P., Zhang, L., Gulla, J.A.: Pre-train, prompt, and recommendation: a comprehensive survey of language modeling paradigm adaptations in recommender systems. Trans. Assoc. Comput. Linguistics **11**, 1553–1571 (2023)
24. Liu, Q., Chen, N., Sakai, T., Wu, X.M.: Once: boosting content-based recommendation with both open-and closed-source large language models. In: Proceedings of the 17th ACM International Conference on Web Search and Data Mining, pp. 452–461 (2024)
25. Liu, Z., Liu, Q., Chen, J., Huang, S., Tang, J., Luo, W.: PYKT: a python library to benchmark deep learning based knowledge tracing models. Adv. Neural. Inf. Process. Syst. **35**, 18542–18555 (2022)
26. Lops, P., De Gemmis, M., Semeraro, G.: Content-based recommender systems: state of the art and trends. In: Recommender Systems Handbook, pp. 73–105 (2011)
27. Ma, Y., Liu, B., Huang, W., Dan, F.: Knowledge graph based recommendation algorithm for educational resource. In: Proceedings of the 14th International Conference on Education Technology and Computers, pp. 436–441 (2022)
28. Moon, T.: The expectation-maximization algorithm. IEEE Signal Process. Mag. **13**(6), 47–60 (1996). https://doi.org/10.1109/79.543975
29. Ren, X., et al.: Representation learning with large language models for recommendation. In: Proceedings of the ACM Web Conference 2024, pp. 3464–3475 (2024)
30. Takami, K., Dai, Y., Flanagan, B., Ogata, H.: Educational explainable recommender usage and its effectiveness in high school summer vacation assignment. In: LAK22: 12th International Learning Analytics and Knowledge Conference, pp. 458–464 (2022)
31. Thaker, K., Barria-Pineda, J.: Using knowledge graph for explainable recommendation of external content in electronic textbooks (2020)
32. Thaker, K., Zhang, L., He, D., Brusilovsky, P.: Recommending remedial readings using student knowledge state. In: International Educational Data Mining Society (2020)
33. Van Meteren, R., Van Someren, M.: Using content-based filtering for recommendation. In: Proceedings of the Machine Learning in the New Information Age: MLnet/ECML2000 Workshop, vol. 30, pp. 47–56. Barcelona (2000)
34. Wei, W., et al.: LLMRec: large language models with graph augmentation for recommendation. In: Proceedings of the 17th ACM International Conference on Web Search and Data Mining, pp. 806–815 (2024)
35. Yudelson, M.V., Koedinger, K.R., Gordon, G.J.: Individualized Bayesian knowledge tracing models. In: Lane, H.C., Yacef, K., Mostow, J., Pavlik, P. (eds.) AIED 2013. LNCS (LNAI), vol. 7926, pp. 171–180. Springer, Heidelberg (2013). https://doi.org/10.1007/978-3-642-39112-5_18
36. Zhang, J., Xie, R., Hou, Y., Zhao, X., Lin, L., Wen, J.R.: Recommendation as instruction following: a large language model empowered recommendation approach. ACM Trans. Inf. Syst. (2023)
37. Zhang, Y., Ai, Q., Chen, X., Wang, P.: Learning over knowledge-base embeddings for recommendation. arXiv preprint arXiv:1803.06540 (2018)
38. Zhou, J., Ma, X., Shan, P., Wang, J.: Learning path recommendation using lesson sequence and learning object based on course graph. In: Proceedings of the 13th International Conference on Education Technology and Computers, pp. 7–12 (2021)